# Homework 03
## REAL VARIABLES I
## Fall 2026

**Due:** 2026-09-13 19:00
**Topics:** Generated outer measures, trace $\sigma$-algebras, Borel $\sigma$-algebras, and product $\sigma$-algebras

**Instructions**
Submission is through Gradescope, accessible through Blackboard.
**Grading rubric:** Every problem/subproblem is graded according to this rubric (no intermediate scores):

- **0%**: No submission or no meaningful content.
- **30%**: Attempt at solution, outline of idea.
- **60%**: Serious attempt at solution. Solution fails to address important points.
- **80%**: Solutions with fixable mistakes or incompleteness.
- **100%**: Complete and correct solution.

You may receive up to a 10% penalty for very unclear or messy exposition.
Please start every solution on a separate page. Clearly mark the pages associated to every part. You may submit either computer typeset solutions or handwritten solutions. Handwritten solutions must be *very clearly* legible.
You are encouraged to work in groups. You may submit the entire homework assignment as a group (up to 4 people). If you only did some exercises as a group, instead of a solution you may clearly indicate whose submission I should look at for the solution to the problem.
Indicate any external help/notes/resources you used. You may not use generative AI in any way to help you with the HW.

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**1.** (When a generated outer measure extends its cost)

**1.a** (weight: 1.0) Let $X$ be a set, let $\mathcal{A} \subseteq 2^X$ contain $\emptyset$, and let $\ell\colon \mathcal{A} \to [0,\infty]$ satisfy $\ell(\emptyset) = 0$. Let $\ell^*$ be the outer measure generated by $\ell$. Prove that

$$\ell^*(A) = \ell(A) \qquad \text{for every } A \in \mathcal{A}$$

if and only if $\ell$ is $\sigma$-subadditive: whenever

$$A \subseteq \bigcup_{n\in\mathbb{N}} A_n \qquad \text{with } A, A_n \in \mathcal{A},$$

then

$$\ell(A) \leq \sum_{n\in\mathbb{N}} \ell(A_n).$$

**2.** (Open approximation of Lebesgue outer measure)

**2.a** (weight: 1.0) Prove that for every set $E \subseteq \mathbb{R}$ and every $\varepsilon > 0$, there exists an open set $A \supseteq E$ such that

$$\mathcal{L}^*(A) \leq \mathcal{L}^*(E) + \varepsilon.$$

**3.** (Induced $\sigma$-algebras)

**3.a** (weight: 0.6) Let $(X, \mathcal{E})$ be a measurable space and let $Y \in \mathcal{E}$. Prove that the following families coincide and define a $\sigma$-algebra on $Y$:

$$\{A \in \mathcal{E} : A \subseteq Y\} \qquad \text{and} \qquad \{A \cap Y : A \in \mathcal{E}\}.$$

This $\sigma$-algebra is called the *induced $\sigma$-algebra* of $\mathcal{E}$ on $Y$.

**3.b** (weight: 0.4) Let $(X, d)$ be a metric space and let $E \subseteq X$ inherit the metric from $X$. Prove that $\mathcal{B}(E)$ coincides with the trace of $\mathcal{B}(X)$ on $E$.

**4.** (Rational half-lines generate the Borel $\sigma$-algebra)

**4.a** (weight: 1.0) Prove that $\mathcal{B}(\mathbb{R})$ is generated by each of the following families:

$$\{(-\infty, b] : b \in \mathbb{Q}\} \qquad \text{and} \qquad \{(-\infty, b) : b \in \mathbb{Q}\}.$$

**5.** (Sections of product measurable sets)

**5.a** (weight: 1.0) Let $(X, \mathcal{E})$ and $(Y, \mathcal{F})$ be measurable spaces. A *measurable rectangle* is a set of the form $A \times B$ with $A \in \mathcal{E}$ and $B \in \mathcal{F}$. Let $\mathcal{E} \otimes \mathcal{F}$ be the $\sigma$-algebra on $X \times Y$ generated by the measurable rectangles. For $S \subseteq X \times Y$, $x \in X$, and $y \in Y$, define the coordinate sections

$$S^y \coloneqq \{x' \in X : (x', y) \in S\} \qquad \text{and} \qquad S_x \coloneqq \{y' \in Y : (x, y') \in S\}.$$

Prove that if $S \in \mathcal{E} \otimes \mathcal{F}$, then $S^y \in \mathcal{E}$ for every $y \in Y$ and $S_x \in \mathcal{F}$ for every $x \in X$.